

1. [20 points] Constructing broadcast towers to cover a highway.

Input: The highway has length $\ell > 0$: starting at position 0 and ending ℓ km away. There are n candidate towers that can be built. Each candidate tower $i \in \{1, \dots, n\}$ has an interval $[a_i, b_i]$ ($a_i < b_i$) such that if the tower is built, the highway is covered from and including position a_i up to and including position b_i .

Objective: Select a smallest set of towers so that every position on the highway is covered by some tower. You may assume that every position on the highway can be covered by some candidate tower, i.e., for every value $x \in [0, \ell]$ there is some tower i such that $x \in [a_i, b_i]$.

- (a) [2 points] Consider the example $[0,5], [3,8], [2,9], [0,3], [4,10], [7,10]$ where $\ell = 10$. Provide an optimal solution.

- (b) [7 points] Provide a greedy rule to optimally solve this problem. Make sure to clearly define the partial solution and how to extend it according to your greedy rule.

Hint: think about how to implement the rule; this will tell you how you should define your partial solutions!

- (c) [8 points] Prove that your greedy rule is correct.

- (d) [3 points] Given a partial solution as you have defined it, what is the runtime needed to find the next tower to add according to your greedy rule? Clearly justify your answer.

2. [10 points] Recall that Karatsuba's algorithm multiplies two n -bit integers using three multiplications of $n/2$ -bit integers, plus some addition taking linear time. This gives a recurrence of $T(n) = 3T(n/2) + n$, and hence an overall time complexity of $\Theta(n^{\log_2 3}) \approx \Theta(n^{1.585})$.

I am considering two alternative algorithms for integer multiplication.

- (a) [4 points] Algorithm A multiplies two n -bit integers using 7 multiplications of $n/4$ -bit integers, plus some addition taking linear time. Write down the recurrence relation for this algorithm, and solve it using the Master Theorem to find the overall asymptotic time complexity.

- (b) [4 points] Algorithm B multiplies two n -bit integers using 5 multiplications of $n/5$ -bit integers, plus sorting an n -element list using a $\Theta(n \log n)$ sorting algorithm. Write down the recurrence relation for this algorithm, and solve it using the Master Theorem to find the overall asymptotic time complexity.

- (c) [2 point] Which of these three algorithms should I use to multiply extremely large numbers? Briefly justify your answer.